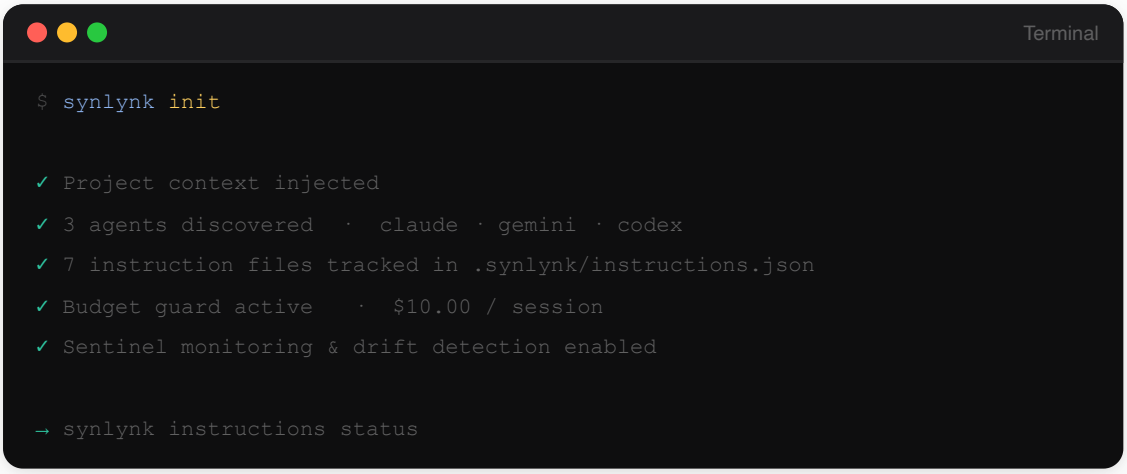
THE OS FOR MULTI-AGENT DEVELOPMENT

synlynk quick start.

From zero to a context-aware, cost-tracked, multi-agent workgroup in under two minutes.

v0.4.1 Instruction Reach

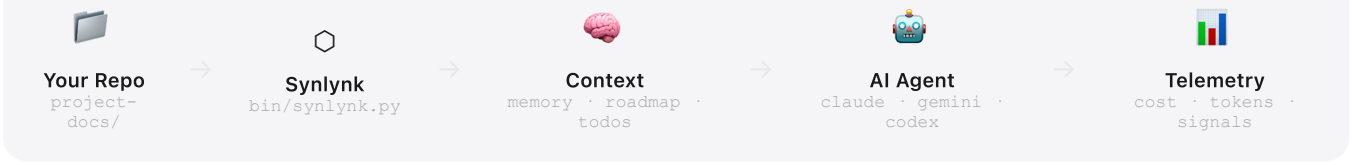


Zero dependencies · Python 3 stdlib only · Claude · Gemini · Codex · Cost tracking · Sentinel guard · Background dispatch

WHAT IS SYNLYNK?

Your AI agents, working together.

Synlynk is a single-file Python CLI that wraps AI agents — Claude, Gemini, Codex — and makes them project-aware, cost-tracked, and safe to run in parallel. One file. Zero dependencies. Works anywhere Python 3 runs.



Why Synlynk?

Context-Aware by Default

Every agent invocation automatically receives your roadmap, memory, and todos. No more copy-pasting context into every session.

Zero Surprise Bills

Budget guard blocks execution the moment you hit your daily or session limit. Real-time cost tracking on every exec.

Hallucination Sentinel

Automatically detects flatline loops, success loops, and quota exhaustion. Alerts before damage is done.

Parallel Dispatch

Launch Claude, Gemini, and Codex simultaneously on different tasks. Monitor all jobs from one CLI while your terminal stays free.

Team-Ready

Single or team mode. Per-user devlogs with git attribution. Shared roadmap and memory visible to every agent in the repo.

Zero Dependencies

Pure Python 3 stdlib. No pip install, no Docker, no build step. One file — works in any environment where Python 3 runs.

1	3	0	∞
FILE — NO BUILD STEP	AI AGENTS SUPPORTED	PIP DEPENDENCIES	PARALLEL AGENT JOBS

GETTING STARTED

Up and running in 60 seconds.

1 **Clone & install globally**

Adds `synlynk` to your PATH via `~/.synlynk/bin/`. No sudo, no pip, no build step required.

install

```
$ git clone https://github.com/nikhilsoman/synlynk
$ cd synlynk && ./install.sh

✓ Installed to ~/.synlynk/bin/synlynk
✓ PATH updated in ~/.zshrc
✓ synlynk 0.4.1 ready
```

2 **Bootstrap any project**

Run inside any repo. Creates `project-docs/` and writes `CLAUDE.md`, `GEMINI.md`, `AI_INSTRUCTIONS.md`. Safe to re-run — skips existing files.

synlynk init

```
~/my-project $ synlynk init

Mode: solo | team? solo

✓ project-docs/roadmap.md · memory.md · todo.md · costs.md
✓ CLAUDE.md · GEMINI.md · AGENTS.md · AI_INSTRUCTIONS.md
✓ Tracked 7 instruction files in .synlynk/instructions.json
✓ Agents: claude · gemini · codex

→ synlynk instructions status
```

3 **Run your first context-injected session**

Synlynk prepends your project context — the AI already knows your roadmap, decisions, and open tasks before you type a word.

synlynk exec

```
$ synlynk exec claude

📋 Context injected   (memory · roadmap · todos)
💰 Budget: $2.34 / $10.00 used this session
🚀 Launching claude...

claude › I can see you're building a billing API for the
monetization module. Based on your roadmap...
```

Magic Moments

The init wizard scans your README, git history, and existing docs to pre-fill your roadmap and memory with real context — not generic placeholders.

V0.4.1 · INSTRUCTION REACH

Run agents in parallel. In the background.

Dispatch fires an agent as a background job — captured to a log file, tracked by the job store. Multiple agents run simultaneously while your terminal stays free.

synlynk dispatch

```
$ synlynk dispatch claude --task "Write unit tests for the billing module"
✓ Dispatched job-a3f2  (claude)  running in background

$ synlynk dispatch gemini --task "Review billing API for edge cases"
✓ Dispatched job-b7e1  (gemini)  running in background

# Both agents working simultaneously — terminal is yours
```

Monitor all jobs

synlynk jobs

```
$ synlynk jobs

• job-a3f2  claude  running 2m14s
• job-b7e1  gemini  running 1m58s
```

Tail a job's output

synlynk logs

```
$ synlynk logs job-a3f2

Writing test_billing.py...
✓ test_charge_success
✓ test_refund_partial
→ 12 / 12 tests passing
```

The Trio Pipeline v0.4.1

Three agents in sequence — Architect designs, Builder implements, Verifier reviews — with full context hand-off between each phase.

synlynk run --trio

```
$ synlynk run --trio "Add Stripe webhook handler"

Phase 1  Architect (claude)  …  ✓ done  job-c1a0
Phase 2  Builder   (gemini)  …  ✓ done  job-d4b2
Phase 3  Verifier  (claude)  …  ✓ done  job-e9f3

✓ Trio complete.  synlynk logs job-e9f3
```

- **Coming in v0.5.0 — Smart Routing**
  Dispatch will automatically route to the best-performing agent for each task domain, based on historical capability scores. Pass `--story <id>` now to start tracking work items.

SENTINEL & BUDGET GUARD

Your safety net, always on.

Synlynk watches every agent invocation for danger signals and surfaces alerts before they become expensive mistakes.

Budget Guard

Set spending limits in `.synlynk/config.json`. Synlynk blocks the next exec the moment you'd exceed your limit. Cost shown after every run.

.synlynk/config.json

```
{
  "limit_usd": 10.00,
  "limit_requests": 100
}
```

budget pulse

```
── Budget Pulse ──────────────
Cost:     $2.34 / $10.00
Requests: 14 / 100
Tokens:   2,840 in · 480 out
```

Sentinel Patterns

Sentinel watches your telemetry log and fires alerts automatically for five danger patterns.

FLATLINE **Same command failed 3× in a row**
Possible hallucination loop. Intervene before more tokens burn.

LOOP **Same command succeeded 5× in <10 min**
Likely an automated loop running unproductively. Investigate.

QUOTA **Quota exhaustion detected in output**
Output matched a known quota-exhausted phrase. Switch agent.

ZOMBIE **Job PID dead or stalled >30 min**
Reconciled automatically on the next run.

DRIFT **Instruction file edited outside synlynk**
SHA mismatch on tracked files. Use `instructions diff/update`.

Managing Alerts

sentinel commands

```
$ synlynk sentinel list

[CRITICAL]  FLATLINE      `synlynk exec claude` failed 3x  — 2026-06-14 09:41
[WARN]    SUCCESS_LOOP  Same command 5x in 4.2 min     — 2026-06-14 10:02

$ synlynk sentinel clear --code FLATLINE
✓ Cleared FLATLINE alerts
```

ALL COMMANDS

Everything at a glance.

Setup & Context

COMMAND	WHAT IT DOES
`synlynk init`	Bootstrap project-docs/, all agent files, and budget config. Safe to re-run.
`synlynk exec <agent>`	Run agent with context injection. Captures token counts and cost.
`synlynk checkpoint`	Regenerate `.synlynk/context.md` from current project-docs.
`synlynk status`	Dashboard: costs, active alerts, running jobs.
`synlynk upgrade`	Check for and apply CLI updates.

Sentinel

COMMAND	WHAT IT DOES
`synlynk sentinel list`	Show all active alerts with severity, code, and timestamp.
`synlynk sentinel clear`	Clear alerts. Target with `--severity` or `--code`.

Dispatch & Jobs v0.4.0

COMMAND	WHAT IT DOES
`synlynk dispatch <agent>`	Fire background job. Accepts `--task` and `--story`.
`synlynk jobs`	List jobs (running / completed / failed). `--all` includes history.
`synlynk logs <job-id>`	Tail job output. Use `--tail N` to limit lines.
`synlynk run --trio`	Architect → Builder → Verifier pipeline with context hand-off.

Instructions v0.4.1

COMMAND	WHAT IT DOES
`instructions status`	Show status of all tracked files (ok / missing / drifted).
`instructions diff [file]`	Show user edits outside the synlynk section.
`instructions update [file]`	Regenerate synlynk section and reset SHA in manifest.
`instructions ack <file>`	Dismiss DRIFT alert without regenerating.

project-docs/	roadmap.md · memory.md · todo.md · costs.md · devlogs/<user>.md
.synlynk/	config.json · context.md · instructions.json · telemetry.json · logs/
state.db	~/.synlynk/projects/<git-root-hash>/state.db (shared across worktrees)

WHAT'S COMING NEXT

The road to v1.0.

Synlynk evolves from an agent wrapper into a full OS for multi-agent development. Each release adds a new layer of the stack.

v0.5 **Capability Engine** ✓ Live · June 2026
Model-aware routing · SQLite WAL state ledger · capability rating & story management

v0.6 **Job Control + Constraints** ✓ Live · June 2026
Constraint propagation · job state machine · PR validation check · model attestation

v0.7 **Async Pipeline + Daemon** Planned · Oct 2026
Background daemon · HTTP context server · synlynk review TUI

v0.8 **Open Context Protocol** Planned · Nov 2026
context --for · checkpoint --from · MCP server · ecosystem interface

v1.0 **Stable OS + Tokq Ready** Planned · Q1 2027
Frozen CLI · pipx/Homebrew distribution · NATS leaf node schema defined